REPORT

STUDENT INFORMATION CHATBOT SYSTEM

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1. Introduction

The purpose of this project is to develop a Student Information Chatbot System using Python and Tkinter. This system is designed to manage and provide essential information about students, such as personal details, academic performance, courses, assignments, and attendance. The chatbot, which is implemented through a simple GUI, allows users to interact with the system by entering a student ID and receiving information in response.

The chatbot provides the following functionalities:

- **Basic Information:** Displays the student's name, GPA, and attendance percentage.
- **Courses:** Lists all the courses a student is enrolled in.
- **Assignments:** Shows the assignments assigned to the student, including the due dates.
- **GPA:** Displays the GPA of the student.

The system is built using Python, with Tkinter being used to create the graphical user interface (GUI).

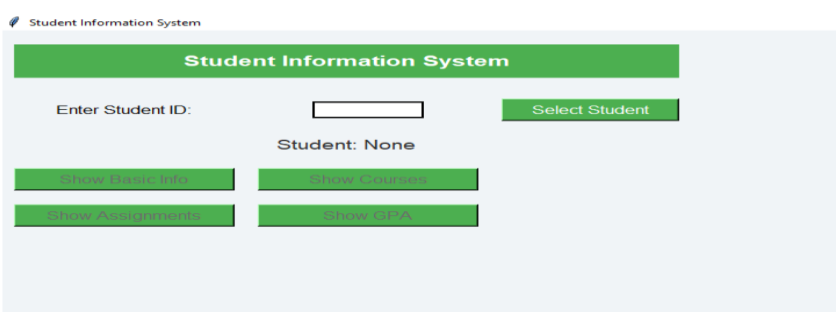
2. System Overview

The Student Information Chatbot System allows for easy access to student data by simply inputting a student ID. The student data is stored as a dictionary, with each student's ID serving as the key. The chatbot system is integrated with Tkinter to present the information in a user-friendly GUI. The system's primary functions are triggered through button clicks, which correspond to the following actions:

1. **Displaying Student Basic Info** (name, GPA, and attendance).
2. **Displaying the Courses** the student is enrolled in.
3. **Displaying the Assignments** the student needs to complete.
4. **Displaying the Student's GPA.**

User Interface

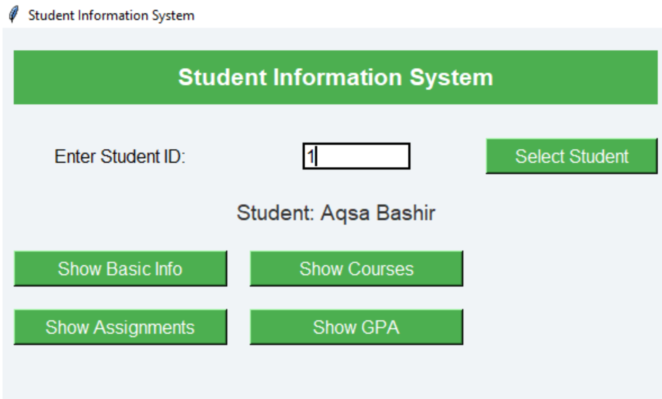
The user interface (UI) is built using Tkinter, a Python library that provides a set of tools to create desktop applications with graphical elements such as labels, buttons, and entry fields.



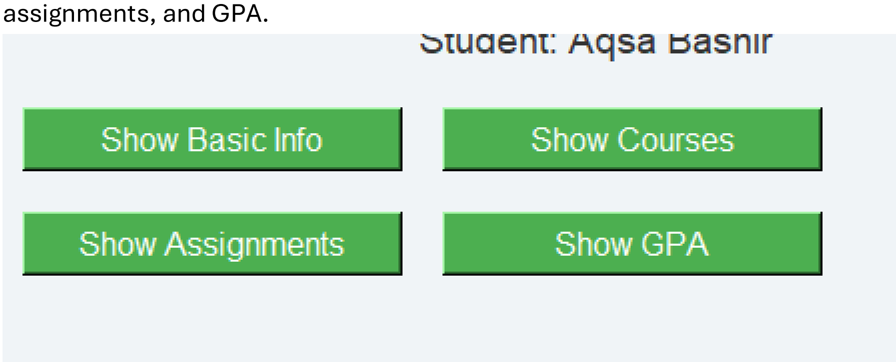
The following components make up the UI:

1. **Student ID Input Field**: The user enters the student ID in this field to select the student whose information they want to access.

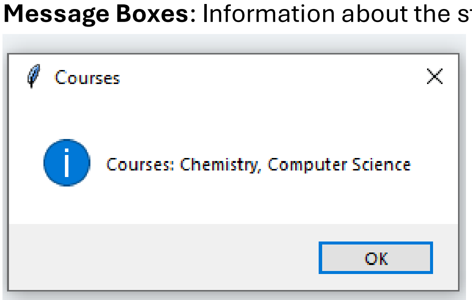
2. **Student Information Display**: The student's name will be shown here after the student ID is entered.



3.
4. **Buttons**: Buttons are provided to allow the user to display the student's basic info, courses, assignments, and GPA.



5. **Message Boxes**: Information about the student is displayed in pop-up message boxes.



Data Storage

The student data is stored in a dictionary named `students_data`. Each student is identified by a unique ID. The dictionary contains the following fields for each student:

- `name`: The student's name.
- `gpa`: The student's GPA.
- `courses`: A list of courses the student is enrolled in.
- `assignments`: A list of assignments with names and due dates.
- `attendance`: The student's attendance percentage.

AI Algorithms

Although the primary focus of this project is to create a simple Student Information Chatbot System, AI algorithms can be integrated to enhance its capabilities. Below are some potential AI algorithms and their applications that could be incorporated into the system in the future.

Natural Language Processing (NLP) for Query Understanding

Currently, the system relies on users inputting a student ID manually. However, integrating **Natural Language Processing (NLP)** can allow the system to understand and process natural language queries. For instance:

- **Query Examples**:
  - "Show me the GPA of Aqsa Bashir."
  - "What assignments does Abdullah have?"
- The chatbot would then parse the user's request, identify relevant information (such as the student's name or the type of data requested), and provide the response accordingly.

Some NLP techniques that could be used:

- **Named Entity Recognition (NER)**: To extract names, course names, or dates from user input.
- **Intent Classification**: To identify whether the user is asking for assignments, GPA, or courses.

Machine Learning for Predictive Analytics

The system could be extended to predict the **academic performance** of students using machine learning algorithms. For instance:

- By analyzing historical data (such as past GPAs, attendance, and assignment completion rates), the system could predict the student's future performance.
- **Supervised learning algorithms**, such as **linear regression** or **decision trees**, could be used to predict student success based on various factors.

Chatbot Enhancement with AI

An AI-powered chatbot could replace the current button-driven interface. The AI model would respond to user questions using a **pre-trained language model**.

- The system could also support **contextual conversations**, allowing users to ask follow-up questions or clarify queries without starting over.

CODE:

```
import tkinter as tk
```

```
from tkinter import messagebox

# Sample data for students
students_data = {
    "1": {
        "name": "Aqsa Bashir",
        "gpa": 3.8,
        "courses": ["English", "Urdu"],
        "assignments": [
            {"name": "English grammar", "due_date": "2024-12-25"},
            {"name": "Urdu nazam", "due_date": "2024-12-26"}
        ],
        "attendance": 95
    },
    "2": {
        "name": "Abdullah",
        "gpa": 3.9,
        "courses": ["Stats", "Data Structures"],
        "assignments": [
            {"name": "Stats presentation", "due_date": "2024-12-24"}
        ],
        "attendance": 90
    },
    "3": {
        "name": "Humaira",
        "gpa": 3.7,
        "courses": ["Chemistry", "Computer Science"],
        "assignments": [
            {"name": "CS Homework", "due_date": "2024-12-20"},
            {"name": "Essay Chem", "due_date": "2024-12-22"}
        ],
        "attendance": 88
    },
    "4": {
        "name": "Ayesha",
        "gpa": 3.5,
        "courses": ["History", "Arts"],
        "assignments": [
            {"name": "Arts gallery", "due_date": "2024-12-23"},
            {"name": "History 1940", "due_date": "2024-12-30"}
        ],
        "attendance": 92
    },
    "5": {
        "name": "Zoha",
```

```
        "gpa": 4.0,
        "courses": ["Biology", "Physics"],
        "assignments": [
            {"name": "Physics Paper", "due_date": "2024-12-28"},
            {"name": "Biology diagrams", "due_date": "2024-12-29"}
        ],
        "attendance": 97
    }
}

# Function to show the basic info of the student
def show_basic_info(student_id):
    student = students_data[student_id]
    info = f"Name: {student['name']}\nGPA: {student['gpa']}\nAttendance: {student['attendance']}%"
    messagebox.showinfo("Basic Info", info)

# Function to show courses of the student
def show_courses(student_id):
    student = students_data[student_id]
    courses = ", ".join(student['courses'])
    messagebox.showinfo("Courses", f"Courses: {courses}")

# Function to show assignments of the student
def show_assignments(student_id):
    student = students_data[student_id]
    assignments = "\n".join([f"- {assignment['name']} (Due: {assignment['due_date']})" for assignment
in student['assignments']])
    messagebox.showinfo("Assignments", f"Assignments:\n{assignments}")

# Function to show GPA of the student
def show_gpa(student_id):
    student = students_data[student_id]
    messagebox.showinfo("GPA", f"GPA: {student['gpa']}")

# Function to handle student selection
def select_student():
    student_id = student_id_entry.get()
    if student_id in students_data:
        display_student_info(student_id)
    else:
        messagebox.showerror("Invalid ID", "Please enter a valid student ID")

# Function to display selected student info
```

```python
def display_student_info(student_id):
    student = students_data[student_id]
    student_name_label.config(text=f"Student: {student['name']}")
    show_info_button.config(state=tk.NORMAL)
    show_courses_button.config(state=tk.NORMAL)
    show_assignments_button.config(state=tk.NORMAL)
    show_gpa_button.config(state=tk.NORMAL)

# Function to handle "Show Info" button click
def show_info():
    student_id = student_id_entry.get()
    show_basic_info(student_id)

# Function to handle "Show Courses" button click
def show_courses_button_click():
    student_id = student_id_entry.get()
    show_courses(student_id)

# Function to handle "Show Assignments" button click
def show_assignments_button_click():
    student_id = student_id_entry.get()
    show_assignments(student_id)

# Function to handle "Show GPA" button click
def show_gpa_button_click():
    student_id = student_id_entry.get()
    show_gpa(student_id)

# Create the main window
window = tk.Tk()
window.title("Student Information System")

# Set window background color and size
window.config(bg="#f0f4f7")
window.geometry("500x400")

# Header Label with title
header_label = tk.Label(window, text="Student Information System", font=("Helvetica", 16, "bold"),
bg="#4CAF50", fg="white", padx=20, pady=10)
header_label.grid(row=0, column=0, columnspan=3, sticky="ew", padx=10, pady=20)
```

```
# Student ID input
student_id_label = tk.Label(window, text="Enter Student ID:", font=("Helvetica", 12), bg="#f0f4f7")
student_id_label.grid(row=1, column=0, padx=10, pady=10)

student_id_entry = tk.Entry(window, font=("Helvetica", 12), width=10, relief="solid", borderwidth=2)
student_id_entry.grid(row=1, column=1, padx=10, pady=10)

# Select Student Button
select_student_button = tk.Button(window, text="Select Student", command=select_student,
bg="#4CAF50", fg="white", font=("Helvetica", 12), relief="raised", padx=20)
select_student_button.grid(row=1, column=2, padx=10, pady=10)

# Student Info display
student_name_label = tk.Label(window, text="Student: None", font=("Helvetica", 14), bg="#f0f4f7",
fg="#333")
student_name_label.grid(row=2, column=0, columnspan=3, padx=10, pady=10)

# Info buttons (initially disabled)
button_bg = "#4CAF50"
button_fg = "white"
show_info_button = tk.Button(window, text="Show Basic Info", command=show_info,
state=tk.DISABLED, bg=button_bg, fg=button_fg, font=("Helvetica", 12), relief="raised", width=20)
show_info_button.grid(row=3, column=0, padx=10, pady=10)

show_courses_button = tk.Button(window, text="Show Courses",
command=show_courses_button_click, state=tk.DISABLED, bg=button_bg, fg=button_fg,
font=("Helvetica", 12), relief="raised", width=20)
show_courses_button.grid(row=3, column=1, padx=10, pady=10)

show_assignments_button = tk.Button(window, text="Show Assignments",
command=show_assignments_button_click, state=tk.DISABLED, bg=button_bg, fg=button_fg,
font=("Helvetica", 12), relief="raised", width=20)
show_assignments_button.grid(row=4, column=0, padx=10, pady=10)

show_gpa_button = tk.Button(window, text="Show GPA", command=show_gpa_button_click,
state=tk.DISABLED, bg=button_bg, fg=button_fg, font=("Helvetica", 12), relief="raised", width=20)
show_gpa_button.grid(row=4, column=1, padx=10, pady=10)

# Start the GUI event loop
window.mainloop()
```

OUTPUT SCREEN